

# Ashkan Pirmani

Applied machine learning · Real world evidence · Drug safety

**NOW** Postdoctoral researcher, KU Leuven and Hasselt University. Innovation Mandate, a Flemish government grant, with Johnson & Johnson Innovative Medicine as industry partner.

**PHD** Dual PhD, 2024. Electrical engineering at KU Leuven, biomedical sciences at Hasselt University.

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|-----------------|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|---------------------------------------------------------------------------|
| <b>EVIDENCE</b> | <b>45.5%</b><br>more patient records in a worldwide COVID-19 study, through the route I built | <b>26,246</b><br>patients in my study of models that adapt to each country | <b>30+</b><br>countries whose registries and consortia I have worked with |
|-----------------|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|---------------------------------------------------------------------------|

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| <b>Experience</b> | 2024 to now | <b>Postdoctoral researcher</b><br>KU Leuven, Department of Electrical Engineering, and Hasselt University, Data Science Institute. Belgium. <ul style="list-style-type: none"><li>• Lead FLkit, an open guide for starting studies across centers that keep their own data. It went from an internal need to a public resource, designed to help teams get started on their own.</li><li>• Wrote the Innovation Mandate proposal and defended it. Awarded in 2026, with Johnson &amp; Johnson Innovative Medicine as industry partner.</li><li>• Took the disability progression study to publication as first author, in npj Digital Medicine (2025).</li><li>• Designed FLoRank, a way for each site to keep a small private part of a shared model. Prototype. Awarded an ECTRIMS personal grant in 2025.</li><li>• Retired FL4E and wrote up why it was not reused, for Medical Informatics Europe 2026.</li><li>• Organizing team of OHDSI Europe 2025. Invited talks at Roche, the Flower community and the North American Society of Artificial Intelligence in Multiple Sclerosis.</li><li>• Coordinate with data holders, clinicians and legal teams in more than thirty countries. Supervise master's students.</li></ul> |
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2020 to 2024

**Doctoral researcher**

KU Leuven and Hasselt University.

- Built the federated route into the worldwide multiple sclerosis and COVID-19 data collection, so registries that could not send data could still take part. It added 3,527 records. First author, JMIR Medical Informatics (2023).
- Built FL4E, a framework for federated studies, and the degree of federation. First author, JMIR Formative Research (2024).
- Ran the first systematic comparison of personalized federated learning for two year disability progression in multiple sclerosis, on 26,246 patients.
- Taught in the Data Science in Healthcare course at Hasselt University, and began supervising master's theses in 2021.
- Presented at ECTRIMS and Medical Informatics Europe in 2024.

2017 to 2019

**Research assistant, Decision Support Systems Lab**

IranDoc, Iranian Research Institute for Information Science and Technology, Tehran.

- Agent based and system dynamics simulation of the national parcel network, searched with a genetic algorithm for a better set of hubs. Best scenario: 15% lower projected transport cost, in simulation.
- Delivered as a decision support dashboard and a policy analysis for senior executives.

**Education**

2020 to 2024

**Dual PhD**

Electrical engineering, KU Leuven. Biomedical sciences, Hasselt University. Defended November 2024.

- Thesis: From Centralized to Federated. The Journey of Data in Healthcare.
- Supervisors: Yves Moreau (KU Leuven), Liesbet M. Peeters (Hasselt University). Co-supervisor: Niels Hellings.

2016 to 2019

**MSc, Industrial Engineering**

Socio-economic systems. Kharazmi University, Tehran.

2015 to 2017

**MBA, Quality Engineering**

Tose'e Institute, Tehran.

**Recognition**

2026

**Innovation Mandate**

Flanders Innovation and Entrepreneurship, with Johnson & Johnson Innovative Medicine as the industry partner. Drug safety research on health records and insurance claims.

|      |                                                                                                          |
|------|----------------------------------------------------------------------------------------------------------|
| 2025 | <b>ECTRIMS personal grant for scientific quality</b><br>For FLoRank, at ECTRIMS 2025, Barcelona.         |
| 2019 | <b>Fully funded PhD scholarship</b><br>KU Leuven and Hasselt University, through to the defense in 2024. |

**Selected contributions****45.5% more records**

Designed and built the federated route into a worldwide COVID-19 and multiple sclerosis study. The four registries counted in the paper added 3,527 records to the 7,757 already collected, and a fifth joined later. Analyses of the combined data set informed worldwide COVID-19 advice for people with multiple sclerosis.

JMIR Medical Informatics 11(1):e48030, 2023

**Personalized to each country**

Personalizing a federated model to each country improved AUC-PR by 31% over the same method without personalization, and ROC-AUC by 3.8% over one model trained on all records pooled together. The best model uses AdaptiveDualBranchNet, an architecture I designed. Simulated federation on records from 26,246 patients and 146 centers. First author, in a collaboration of 73.

npj Digital Medicine 8(478), 2025

**The degree of federation**

Each center chooses whether its records leave. On a public benchmark, letting half the centers send their records kept 97.5% of the fully federated score.

JMIR Formative Research 8:e55496, 2024

**A public resource, maintained**

FLkit: 39 pages, eleven ways in by role, seven worked project stories, 29 contributors. Supported by ELIXIR Belgium. Its use has not been evaluated yet.

arXiv 2606.23500, 2026

**Teaching and supervision**

2022 to 2024

**Data Science in Healthcare**

Teaching member, Healthcare Engineering program, Hasselt University. Two academic years.

2021 to now

**Master's thesis supervision**

Seven master's theses in artificial intelligence, KU Leuven.

- Predicting admission to care. Federated, pooled and local data compared.
- How splitting the data changes what the model learns.
- Federated boosting in semi-supervised learning.
- Contrastive learning for federated models.
- Personalized models. Architectures and parameter exchange.
- Fine-tuning a global federated model for local performance.
- Early disease detection from shallow whole genome sequencing of cell-free DNA.

**Methods and systems****Methods**

- Federated learning and federated analysis
- Personalization when sites hold different patients
- Privacy-preserving machine learning
- Survival analysis and classification on clinical data

**Data**

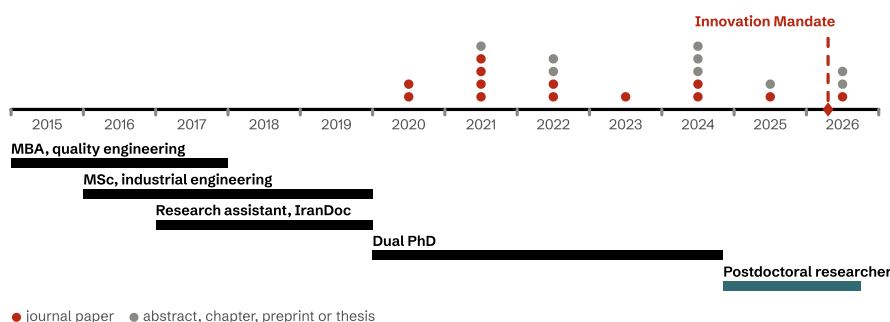
- Real world and observational data
- Cohort building and phenotype definition across registries
- Registry harmonization and data quality
- Data feasibility: whether a question can be answered at all
- OMOP common data model, through OHDSI
- FAIR data practice

**Tools**

- Python, PyTorch, Flower, scikit-learn, pandas
- R and SQL
- Docker, Kubernetes, continuous integration

**Networks**

- OHDSI Belgium, and the organizing team of OHDSI Europe 2025
- ELIXIR Belgium
- The MSBase registry and the MS Data Alliance

**Selected publications**

● journal paper    ● abstract, chapter, preprint or thesis

Roles as bars, and every entry in the publications list as a dot in its year.

**2026****Development and Design of FLKit: A Structured Onboarding Toolkit for Federated Learning in Health and Life Sciences**

*arXiv preprint arXiv:2606.23500*. Pirmani, Vermeulen, Vinterhalter, Geys, Faes, Ali, et al.

**2026****Good for All, Not Good Enough for One: Reuse Dilemma in Federated Learning***Studies in Health Technology and Informatics*. Pirmani, Moreau, Peeters**2025****FLoRank: adaptive structured personalisation for federated prediction of disability progression in multiple sclerosis***ECTRIMS 2025, Multiple Sclerosis Journal 31(3 suppl)*. Pirmani, Moreau, Peeters**2025****Personalized federated learning for predicting disability progression in multiple sclerosis using real-world routine clinical data***npj Digital Medicine*. Pirmani, De Brouwer, Arany, Oldenhof, Passemiers, Faes, et al.**2024****Accessible ecosystem for clinical research (federated learning for everyone): development and usability study***JMIR Formative Research*. Pirmani, Oldenhof, Peeters, De Brouwer, Moreau**2023****The Journey of Data Within a Global Data Sharing Initiative: A Federated 3-Layer Data Analysis Pipeline to Scale Up Multiple Sclerosis Research***JMIR Medical Informatics*. Pirmani, De Brouwer, Geys, Parciak, Moreau, Peeters**2022****Updated results of the COVID-19 in MS global data sharing initiative: anti-CD20 and other risk factors associated with COVID-19 severity***Neurology: Neuroimmunology & Neuroinflammation*. Simpson-Yap, Pirmani, Kalincik, De Brouwer, Geys, Parciak, et al.**2021****Associations of disease-modifying therapies with COVID-19 severity in multiple sclerosis***Neurology*. Simpson-Yap, De Brouwer, Kalincik, Rijke, Hillert, Walton, et al.